一，求基本初等函数的导数

(1) C(常数)

(2) x^k

(3) $\sin x$

(4) $\cos x$

(5) $\tan x$

(6) $\cot x$

(7) $\sec x$

(8) $\csc x$

(9) a^x

(10) e^x

(11) $\log_a x$

(12) $\ln x$

(13) $\arcsin x$

(14) $\arccos x$

(15) $\arctan x$

(16) $\operatorname{arccot} x$

二，求下列函数的微分

(1) $y = \frac{1}{x} + 2\sqrt{x}$

(2) $y = x\sin(2x)$

(3) $y = \frac{x}{\sqrt{x^2+1}}$

(4) $y = \ln^2(1-x)$

(5) $y = x^2e^{2x}$

(6) $y = e^{-x}\cos(3-x)$

(7) $y = \arcsin\sqrt{1-x^2}$

(8) $y = \tan^2(1+2x^2)$

(9) $y = arctan\frac{1-x^2}{1+x}$

(10) $s = A\sin(\omega t + \varphi)$ $(A, \omega, \varphi$是常数)

(11) $y = x\arcsin(x) + \sin(\tan x)$

(12) $y = \ln(\sin x)$

(13) $y = \ln(x+a)$

(14) $y = x^2e^x$

(15) $y = x^2\ln x\cos x$

(16) $y = \sqrt{x^2-1}$

(17) $y = x^2 e^{-ax^2}$

(18) $y = xe^{-ax^2}$

(19) $y = x^2 e^{-ax}$

(20) $y = xtanx$

(21) $y = 5 + 3x - 4x^3$

(22) $y = 3x^4 - 2x^2 + 8$

(23) $y = 2x^3 - 6x^2 + 7x$

(24) $y = e^{1-3x} cosx$

(25) $y = \mathrm{In}\left(1 + e^{x^2}\right)$

(26) $y = (x-a)(x-b)(x-c)$

(27) $\mathrm{y} = \mathrm{xlnx} - \frac{1}{2}x^2 + x$

(28) $\mathrm{y} = x^3 + 3x^2 + 3x - 9$

(29) $\mathrm{y} = 5x^3 - 2^x + 3e^x$

(30) $\mathrm{y} = 2\mathrm{tanx} + \mathrm{secx} - 1$

(31) $\mathrm{y} = x^2 lnx$

(32) $\mathrm{y} = \frac{lnx}{x}$

(33) $\mathrm{y} = \frac{e^x}{x^2} + ln3$

(34) $\mathrm{y} = \frac{1+sint}{1+cost}$